

Python Course for Physicists

Chunk 2: Variables

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08.10.2025

Live Coding: Text Output – Printing

- Print Statement:

```
print("Hello World")
```



```
print("Hello World")
```

```
Hello World
```

- Print Numbers:

```
print(42)
```



```
print(42)
```

```
42
```

- Saving Data into Memory:
Variables

```
a = "Hello World"
```

```
print(a)
```

```
Hello World
```

Live Coding: Comments

- Comments don't get executed → Ignored by the program

- Single line comment: `#`

Can be at end of line (inline)

```
# comment before  
print("Hello World") # inline comment  
# comment after
```

```
Hello World
```

- Multiline comment: `""" ... """`

```
"""  
This is a  
really long  
multiline  
comment  
"""  
print("Hello World")
```

```
Hello World
```

Variables Names:

- Usually alphanumeric + “_” → `a`, `tree`, `house99`, `tree_house1`.
 - First character cannot be number → Forbidden: `1house`

- Reserved Keywords (don't use):

According to ChatGPT:

False, None, True, and, as, assert, async, await, break, case, class, continue, def, del, elif, else, except, finally, for, from, global, if, import, in, is, lambda, match, nonlocal, not, or, pass, raise, return, try, while, with, yield

- DON'T:

```
print = 5
print("Hi")
```

```
-----
TypeError
Cell In[8], line 2
      1 print = 5
----> 2 print("Hi")
```

```
TypeError: 'int' object is not callable
```

What's the difference: "Hello World" & 42

- → Different “types” of Data

Name	What	Example	
String	Text	"Hello World"	Single or Double Quotes
int	Integer (Number)	1 , 10 , -123	
float	Rational Number	3.14 , 1.99 , 1.2e-6	Btw. complex numbers are noted by “j”: X = 4j
boolean	True / False	True , False	
Emptiness	Explicitly saving nothing	None	

What's the difference: "Hello World" & 42

```
this_is_a_string = "Hello Mars"  
this_is_a_int = 10  
this_is_a_float = 9.31  
this_is_a_boolean = True  
this_is_no_value = None  
print(this_is_a_string)  
print(this_is_a_int)  
print(this_is_a_float)  
print(this_is_a_boolean)  
print(this_is_no_value)  
print()  
print(this_is_a_string, this_is_a_int)
```



```
Hello Mars  
10  
9.31  
True  
None  
  
Hello Mars 10
```



Separating print arguments by “,” prints them with a Whitespace inbetween

Reassignment / Memory / Restarting Kernel

- Reassignment:

```
b = 1
print(b)
b = "I am now a beautiful string"
print(b)
```

1
I am now a beautiful string

- Memory:

```
a = 1
print(a)
```

1



```
print(a)
```

1

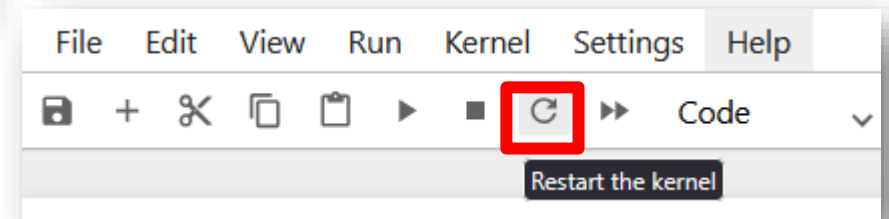


- Restarting the kernel → Delete Memory:

```
print(a)
```

NameError Traceback (most recent call last)
Cell In[1], line 1
----> 1 print(a)

NameError: name 'a' is not defined



Live Coding: Arithmetic Operators

- Example with Integers:
→ Adding to itself possible

```
a = 2
b = 5
c = a + b
print(c)
```

7

```
q = 10
print(q)
q = q + 5
print(q)
```

10
15

- Example with Strings:

```
a = "Hello"
b = "World"
c = a + b
print(c)
```

HelloWorld

- Multiplying a String:

```
a = "Hello"
print(3*a)
```

HelloHelloHello

Arithmetic Operators: Complete List

Data Type	+	-	*	/	// (Floor Divide)	% (Modulo)	** (Exponentiate)
float	Valid	Valid	Valid	Valid	Valid (floored float)	Valid	Valid
int	Valid	Valid	Valid	Valid (float result)	Valid (int result)	Valid	Valid
string	Concatenation	Invalid	Repeat (with int)	Invalid	Invalid	Invalid	Invalid
boolean	Treated as int (0,1)	Treated as int	Treated as int	Treated as int	Treated as int	Treated as int	Treated as int

Type Conversion: Type Parsing

- Determine type of (unknown) variable:

```
print(type(a))
```

```
<class 'str'>
```

- Converting the Type:

```
a = "1"  
print(a)  
print(int(a))  
print(float(a))  
print(bool(a))  
print(str(a))
```



```
1  
1  
1.0  
True  
1
```

Live Coding: Type Conversion: Type Parsing

- Why is this useful?
→ Concatenate string and int:

```
number = 5  
print("A nice number: " + number)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[15], line 2  
      1 number = 5  
----> 2 print("A nice number: " + number)  
  
TypeError: can only concatenate str (not "int") to str
```

- Type Parsing int → str fixes the problem:

```
number = 5  
print("A nice number: " + str(number))  
  
A nice number: 5
```

- A lot of type parsing happens automatically:

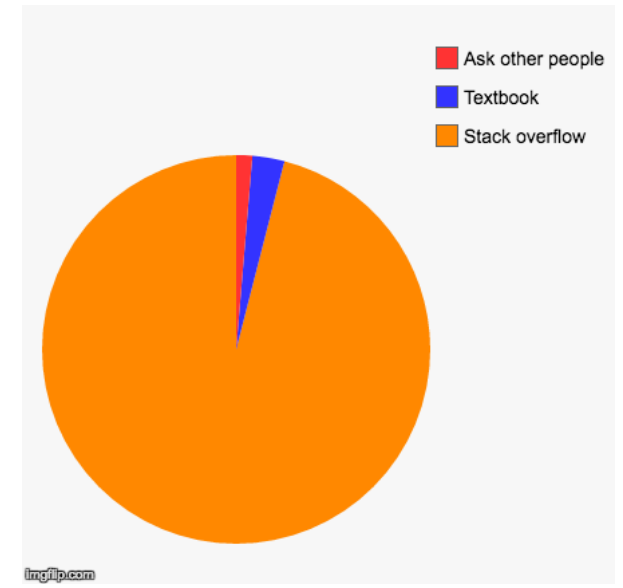
```
integer = 1  
print(type(integer))  
integer = integer / 2  
print(type(integer))
```

```
<class 'int'>  
<class 'float'>
```

Some notes on solving task sheets

- Splitting into second room
- If you get stuck:
 - Try yourself again
 - Ask your neighbor
 - Ask assistants
 - If busy: Ask google (Stackoverflow <3)
 - (Maybe) ask LLM (ChatGPT / Perplexity / Copilot)
 - Look into solution file
- Struggling and debugging are 90% of a coders work

Good Luck 😊



[Deleted User on Reddit]



[Programmerhumor.io]